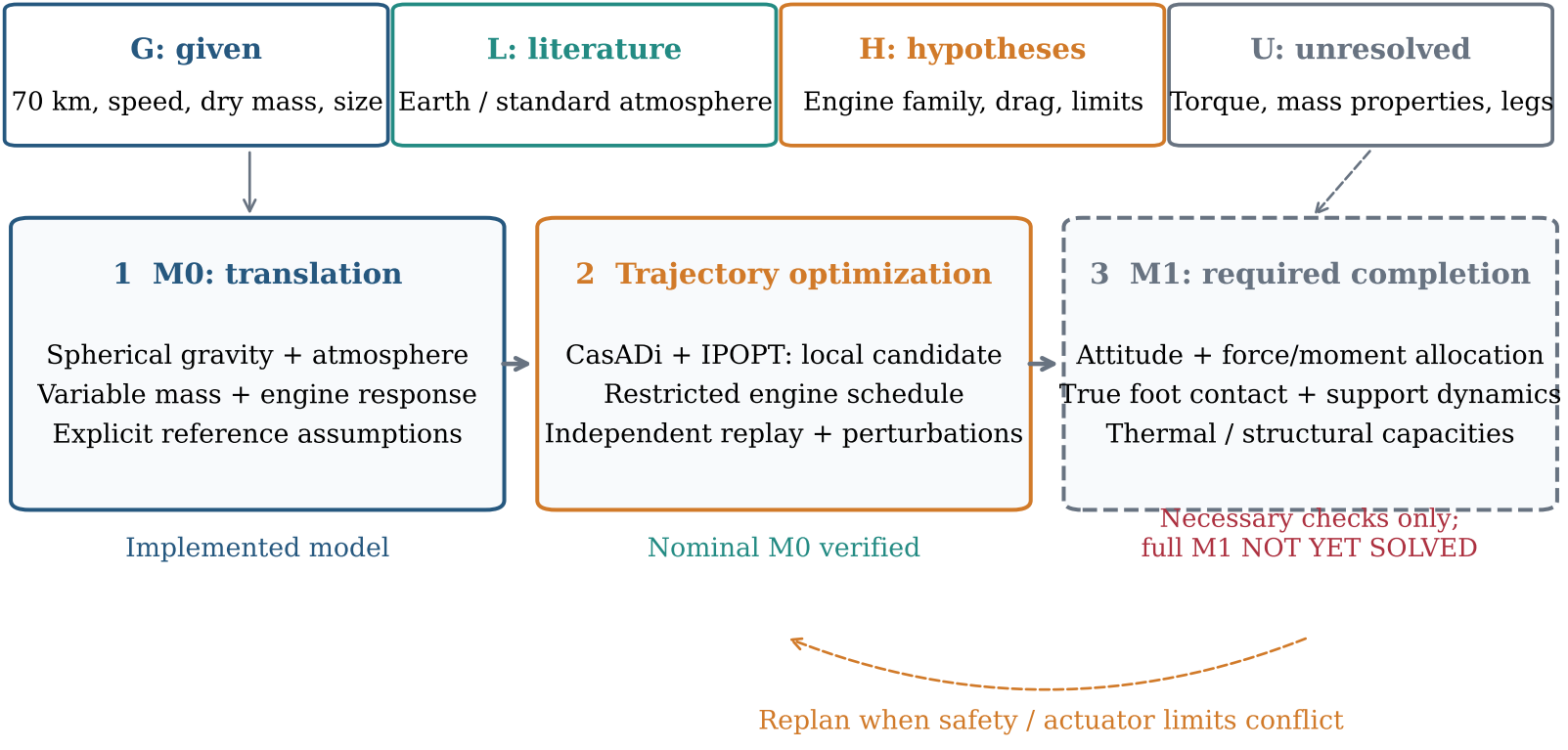
G: given
70 km, speed, dry mass, size
L: literature
Earth / standard atmosphere
H: hypotheses
Engine family, drag, limits
U: unresolved
Torque, mass properties, legs
1 M0: translation
Spherical gravity + atmosphere
Variable mass + engine response
Explicit reference assumptions
2 Trajectory optimization
CasADi + IPOPT: local candidate
Restricted engine schedule
Independent replay + perturbations
3 M1: required completion
Attitude + force/moment allocation
True foot contact + support dynamics
Thermal / structural capacities
Implemented model
Nominal M0 verified
Necessary checks only;
full M1 NOT YET SOLVED
Replan when safety / actuator limits conflict